

Hierarchical Matrix Approximations in Transformers: From Vision to Language Models

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Abstract—Dense weight matrices in transformer architectures carry substantial computational and memory overhead, much of which may be redundant due to the structured nature of the data being modeled. We investigate *Hierarchical matrices* (H-matrices) as structured replacements for dense projection matrices in transformers, exploiting the observation that attention weight matrices often exhibit hierarchical low-rank structure. We conduct experiments in two phases. First, in a Vision Transformer (ViT) setting, we replace QKV projection matrices with H-matrix approximations and train from scratch on CIFAR-10, demonstrating a 48% reduction in parameters and a 15% reduction in training time per epoch at matrix size $N=4096$, with only an $\approx 2\%$ accuracy drop. Second, we adopt a fine-tuning paradigm for Large Language Models (LLMs), inserting additive H-matrix updates—analogue to LoRA—into the attention layers of TinyLlama-1.1B. We benchmark against LoRA across WikiText-2 language modeling, medical QA (MedQA-USMLE, PubMedQA), and legal QA (CaseHOLD). A 3-epoch medical QA comparison (H-MAT $\ell=1, r=64$ vs. LoRA $r=128$, dropout=0.05, QKV) yields near-identical scores: MedQA 26.24% vs. 26.16%, PubMedQA 63.40% vs. 63.80%. On CaseHOLD, H-MAT achieves 78.7% against LoRA’s 80.2%, both representing gains of over 55 percentage points above the pretrained baseline of 20.4%. Across two specialised domains, H-matrix adapters consistently close to within 1–2% of LoRA at matched parameter counts, motivating further study at larger scale.

Index Terms—Hierarchical matrices, Vision Transformers, LoRA, parameter-efficient fine-tuning, TinyLlama, attention mechanism

I. INTRODUCTION

Transformers [1] power both vision [2] and language models [3]. A key computational bottleneck is the dense matrix multiplication in multi-head self-attention (MHSA): the query, key, and value projections $W_Q, W_K, W_V \in \mathbb{R}^{d \times d}$ each require $O(d^2)$ parameters, becoming prohibitive as d scales.

A classical numerical linear algebra tool—the *Hierarchical matrix* (H-matrix)—approximates dense matrices with far fewer degrees of freedom by exploiting low-rank structure in off-diagonal blocks [5], [6]. Elements far from the diagonal interact weakly and can be compressed; those near the diagonal retain full-rank representations. This mirrors the local dependency structure in images: nearby patches couple strongly, distant patches weakly.

This project investigates whether H-matrix structure can be exploited in transformer weight matrices to reduce parameter counts without sacrificing accuracy. We pursue two tracks:

- 1) **ViT from scratch:** Replace dense QKV matrices with H-matrices and train on CIFAR-10.

- 2) **LLM fine-tuning:** First replace the pretrained weights directly with H-matrices (WikiText-2), then pivot to an additive LoRA-style formulation [4] and benchmark on TinyLlama-1.1B across general, medical, and legal NLP tasks.

Section II provides background, Section III covers ViT experiments, Section IV covers LLM fine-tuning, Section V discusses findings, and Section VI concludes.

II. BACKGROUND

A. Transformer Attention

Given token embeddings $X \in \mathbb{R}^{n \times d}$, multi-head self-attention computes:

$$\text{Attn}(X) = \text{softmax}\left(\frac{XW_Q(XW_K)^\top}{\sqrt{d_k}}\right)XW_V, \quad (1)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$ are fully dense, requiring $O(d^2)$ storage and computation per projection.

B. Hierarchical Matrices

An H-matrix $\mathcal{H} \in \mathbb{R}^{N \times N}$ is defined recursively via a *cluster tree*. Diagonal blocks (nearby interactions) are full-rank; off-diagonal blocks (distant interactions) are approximated as $UV^\top$ with $r \ll \min(m, n)$. The level ℓ controls subdivision depth:

$$\mathcal{H}^{(\ell)} = \begin{cases} UV^\top & \ell = 0, \\ \begin{bmatrix} D_1 & U_{12}V_{12}^\top \\ U_{21}V_{21}^\top & D_2 \end{bmatrix} & \ell \geq 1, \end{cases} \quad (2)$$

where D_i are diagonal blocks (full or recursively H-matrix). Parameter count is $O(rN\ell)$ vs. $O(N^2)$ for dense. At $\ell=0$, $\mathcal{H}$ collapses to a LoRA-identical $UV^\top$.

C. Why H-Matrices for Transformers?

Spatially proximate ViT patches share strong correlations; the attention map of a trained ViT exhibits approximate hierarchical low-rank structure [7]. We hypothesise that the weight matrices producing Q, K, V also admit H-matrix approximations. For LLMs, the question is whether H-matrix structure—either as a direct replacement or as an additive update—can match or exceed the parameter efficiency of LoRA.

III. PHASE 1: VISION TRANSFORMER EXPERIMENTS

A. Setup

We used an open-source small-scale ViT and trained two variants from scratch on **CIFAR-10** under identical protocols:

- **Baseline ViT:** Dense QKV projections.
- **H-ViT:** QKV replaced with H-matrix constructions (randomised initialisation).

To investigate the theoretical advantage of H-matrices—which manifests only for sufficiently large N —we ran experiments across multiple values of N . At small N , H-ViT was slower than the dense baseline due to GPU inefficiency with many small matrix multiplications. As N increased, the per-epoch training time of H-ViT decreased relative to the baseline, eventually surpassing it in efficiency. Results at $N=4096$ confirm this crossover.

B. Results

TABLE I

ViT DENSE VS. H-MATRIX QKV AT $N=4096$ ON CIFAR-10. VALUES ARE REDUCTIONS RELATIVE TO THE DENSE BASELINE.

Model	Δ Params	Δ Time/Epoch	Δ Acc.
Dense ViT (baseline)	—	—	—
H-ViT ($N = 4096$)	−48%	−15%	$\approx -2\%$

H-ViT achieves a **48% parameter reduction** and **15% per-epoch speedup** at the cost of only an $\approx 2\%$ accuracy drop—a strong efficiency-accuracy tradeoff.

C. GPU Bottleneck at Small N

The H-matrix decomposition replaces one large matrix multiply with many small ones. Modern GPUs are highly optimised for large batched BLAS operations; this fragmentation causes inefficient memory access and kernel launch overhead, making H-ViT *slower* than the dense baseline at typical hidden dimensions ($d \leq 1024$). The crossover is only reached at large N , confirmed by our $N=4096$ results.

To our knowledge, no public ViT uses projection matrices with $N > 2048$, making fair benchmarking against published models challenging. Proper evaluation requires large-scale data (e.g., ImageNet-1K) at large N , beyond the current compute budget.

IV. PHASE 2: LLM FINE-TUNING

A. Motivation

ViT experiments established proof-of-concept but revealed that large N is essential for practical advantages. LLMs have naturally larger hidden dimensions (TinyLlama-1.1B: $d=2048$), and fine-tuning rather than training from scratch dramatically reduces compute. We explore two fine-tuning strategies in sequence, with the first informing the design of the second.

B. Model and Datasets

Base model: TinyLlama-1.1B (intermediate-step-1431k-3T).

Phase 2a – WikiText-2 [8]: Causal LM objective, block size 512. H-matrix weights *replace* the pretrained projection matrices directly ($W \leftarrow \mathcal{H}$), and are compared against LoRA at matched parameter counts.

Phase 2b – Medical QA: Fine-tuned on MedQA (USMLE) [9]; evaluated on MedQA-4-options and PubMedQA [10]. Motivated by the limitations of direct replacement (Section IV-C), this phase adopts the *additive* formulation $W = W_0 + \Delta W_{\mathcal{H}}$. Loss is computed on the answer token only.

Phase 2c – Legal QA (CaseHOLD): To test whether findings generalise beyond medical QA, we fine-tune on **CaseHOLD** [11] from the LexGLUE benchmark suite [12]. Each example presents a legal citation context and five candidate holdings; the model selects the correct one. We use 5,000 training and 1,000 validation examples, with the same answer-token loss, UV decomposition, and dropout ($p=0.05$) as Phase 2b. Evaluation is on a held-out test split of 1,000 examples using CaseHOLD accuracy.

C. WikiText-2: Direct Replacement

In this initial LLM approach, the pretrained weight matrix is replaced entirely by an H-matrix ($W \leftarrow \mathcal{H}$), which is then fine-tuned. Adapters were applied to W_Q only or to all of QKV, with r and ℓ varied while matching trainable parameter counts between LoRA and H-MAT.

TABLE II

WIKITEXT-2 RESULTS, TINYLLAMA-1.1B ($W \leftarrow \mathcal{H}$). NO FINE-TUNING BASELINE: 49.66%.

Method	Target	Params	Config	Acc. (%)
LoRA	QKV	83.68M	$r = 437$	55.78
H-MAT	QKV	83.79M	$r = 128, \ell = 2$	55.55
LoRA	W_Q only	80.74M	matched to $\ell = 4$	55.39
H-MAT	W_Q only	80.87M	$r = 128, \ell = 4$	55.31
LoRA	W_Q only	55.69M	matched to $\ell = 2$	55.45
H-MAT	W_Q only	55.70M	$r = 128, \ell = 2$	57.26

H-MAT closely matches or exceeds LoRA at matched parameter counts. For QKV and W_Q -only at higher levels ($\ell=4$), LoRA holds a small advantage. However, at W_Q -only with $\ell=2$, H-MAT achieves 57.26% against LoRA’s 55.45%—a notable reversal suggesting the hierarchical block structure can be beneficial at certain configurations. In all cases, replacing pretrained weights entirely discards the rich representations encoded in W_0 , which motivated a pivot to the additive formulation for subsequent phases.

D. Medical QA: Additive H-Matrix Adapters

1) *H-Matrix Adapter Design:* Inspired by LoRA [4], which freezes W_0 and learns a low-rank correction $\Delta W = BA$, we define the *H-Matrix Adapter*:

$$W = W_0 + \Delta W_{\mathcal{H}}, \quad (3)$$

where $\Delta W_{\mathcal{H}}$ is an H-matrix with level ℓ and rank r , and only $\Delta W_{\mathcal{H}}$ is trained. This preserves the pretrained weights and restricts updates to a structured, parameter-efficient subspace.

Initialisation: U sampled from Gaussian, V set to zero, so $\Delta W_{\mathcal{H}}=0$ at the start of training (matching LoRA). **Dropout:** $p=0.05$, matching LoRA’s default regularisation.

Note on baselines: Throughout all QA experiments, LoRA uses its standard $UV^{\top}$ decomposition with dropout $p=0.05$ by default. Decomposition labels (USV, UV) and dropout annotations in the following subsections and table captions describe the *H-MAT* configuration being evaluated, not changes to the LoRA baseline.

2) *Evolution of the Approach:* Even within the additive formulation, the experiments went through further key refinements:

(1) **Answer-token loss masking.** Restricting the causal LM loss to the answer token only—rather than the full prompt and options—improved both LoRA and H-MAT on the QA benchmarks.

(2) **Removing the S factor.** The initial H-MAT used a three-factor decomposition ($USV^{\top}$) for off-diagonal blocks. Since LoRA uses only $UV^{\top}$, we removed S to align both methods parametrically.

(3) **Adding dropout.** Dropout ($p=0.05$) was added to H-MAT to match LoRA’s regularisation exactly.

3) *1-Epoch QKV Results: H-MAT with USV Decomposition:* Table III shows results with the additive update and answer-token loss, using the initial H-MAT USV decomposition (no dropout on H-MAT).

TABLE III

MEDICAL QA — QKV ADAPTERS, 1 EPOCH, ANSWER-TOKEN LOSS.
H-MAT: USV DECOMPOSITION, NO DROPOUT. PRETRAINED: MEDQA 24.74%, PUBMEDQA 62.8%.

Method	Config	MedQA (%)	PubMedQA (%)
Pretrained	—	24.74	62.8
LoRA	$r = 16, 3.06\text{M}$	27.49	62.8
H-MAT	$\ell = 1, r = 8, 3.45\text{M}$	24.74	61.4
LoRA	$r = 128, 24.5\text{M}$	27.65	62.6
H-MAT	$\ell = 1, r = 64$	25.37	60.8
H-MAT	$\ell = 0, r = 128$	24.51	61.0

With USV decomposition and no dropout on H-MAT, it falls short of LoRA. The extra S factor may introduce training instability, and the absence of dropout leaves H-MAT under-regularised relative to LoRA.

4) *1-Epoch QKV Results: H-MAT with UV + Dropout:* After removing S and adding dropout to H-MAT, performance improved markedly (Table IV).

With UV decomposition and dropout on H-MAT, the best H-MAT configuration ($\ell=0, r=128$, dropout=0.05) achieves 27.73% on MedQA and 65.4% on PubMedQA. One-epoch results show high variance with no clear systematic trend across configurations, motivating a longer-training comparison.

5) *3-Epoch Comparison (QKV, Dropout=0.05):* To assess whether trends stabilise, we ran a 3-epoch comparison applying

TABLE IV

MEDICAL QA — QKV ADAPTERS, 1 EPOCH, ANSWER-TOKEN LOSS.
H-MAT: UV DECOMPOSITION, DROPOUT=0.05 WHERE NOTED.

Method	Config	MedQA (%)	PubMedQA (%)
LoRA	$r = 128$, dropout=0 (H-MAT only)	24.35	49.8
H-MAT	$\ell = 0, r = 128$, dropout=0	24.35	64.2
H-MAT	$\ell = 1, r = 64$, dropout=0.05	26.63	64.6
H-MAT	$\ell = 1, r = 8$, dropout=0.05	24.67	62.2
H-MAT	$\ell = 0, r = 128$, dropout=0.05	27.73	65.4

adapters to all three projections (Q, K, V) with H-MAT using UV decomposition and dropout=0.05. Configurations were **LoRA** $r=128$ and **H-MAT** $\ell=1, r=64$, chosen to match parameter counts.

TABLE V

MEDICAL QA — QKV ADAPTERS, 3 EPOCHS, ANSWER-TOKEN LOSS.
H-MAT: UV DECOMPOSITION, DROPOUT=0.05.

Method	Config	MedQA (%)	PubMedQA (%)
H-MAT	$\ell = 1, r = 64$, dropout=0.05	26.24	63.40
LoRA	$r = 128$	26.16	63.80

After 3 epochs, **H-MAT and LoRA converge to near-identical performance.** H-MAT leads marginally on MedQA (26.24% vs. 26.16%) while LoRA leads marginally on PubMedQA (63.80% vs. 63.40%), demonstrating that a well-configured H-matrix adapter fully matches LoRA in the longer-training regime.

E. Legal QA: CaseHOLD

To assess generalisation to a second out-of-domain task, we fine-tuned TinyLlama-1.1B on CaseHOLD under the same additive H-matrix adapter setup used for medical QA (H-MAT: UV decomposition, dropout=0.05, answer-token loss, 3 epochs, QKV projections). The pretrained model scores only 20.4% on CaseHOLD, barely above random chance for a 5-way task (20%), confirming the domain is genuinely out-of-distribution for the base model.

TABLE VI

CASEHOLD ACCURACY — QKV ADAPTERS, 3 EPOCHS, ANSWER-TOKEN LOSS. H-MAT: UV DECOMPOSITION, DROPOUT=0.05. PRETRAINED BASELINE: 20.4%.

Method	Config	CaseHOLD Acc. (%)
Pretrained (no FT)	—	20.4
LoRA	$r = 128$	80.2
H-MAT	$\ell = 1, r = 64$	78.7
H-MAT	$\ell = 2, r = 32$	76.0

Both methods achieve a dramatic improvement over the pretrained baseline—an absolute gain exceeding 55 percentage points—confirming that adapter-based fine-tuning is highly effective even for a model with no prior legal exposure. LoRA ($r=128$) reaches 80.2%, while H-MAT at $\ell=1, r=64$ achieves 78.7%, a gap of 1.5 percentage points at matched parameter

counts. Increasing the hierarchical depth to $\ell=2$ with $r=32$ (further reducing parameters) yields 76%, showing a modest accuracy–efficiency tradeoff as the H-matrix structure becomes more constrained. The trend is consistent with medical QA: H-MAT closes to within $\approx 1\text{--}2\%$ of LoRA on domain-specific tasks while using structured, parameter-efficient updates.

V. DISCUSSION

A. When Do H-Matrices Help?

H-MAT adapters are most competitive with LoRA when:

- Matrix dimension N is large (ViT at $N=4096$), enabling the complexity advantage of H-matrices to manifest.
- The additive formulation is used, preserving pretrained representations rather than replacing them.
- H-MAT uses UV decomposition and dropout ($p=0.05$), aligning its regularisation with LoRA.
- The task is domain-specific (medical QA, legal QA), where H-MAT consistently closes to within $1\text{--}2\%$ of LoRA at matched parameter counts.
- Training is extended to multiple epochs, allowing the hierarchical structure to be learned effectively.

On general LM (WikiText-2), LoRA holds a consistent advantage under direct replacement, likely because globally unconstrained low-rank updates better capture broad language patterns. On specialised tasks—both medical and legal—the gap narrows substantially, suggesting the structured inductive bias of H-matrices is better matched to constrained domain adaptation than to open-ended language modelling.

B. Limitations

- 1) **Scale:** ViT experiments used CIFAR-10; LLM experiments ran at most 3 epochs on a 1.1B model. Larger-scale training is needed for definitive conclusions.
- 2) **GPU efficiency:** H-MAT operations involve many small matrix multiplications that under-utilise GPU BLAS at small N . Custom CUDA kernels could substantially close this gap.
- 3) **Hyperparameter sensitivity:** H-MAT introduces additional hyperparameters (ℓ , block structure) beyond LoRA’s single rank r , and systematic search was limited by compute.
- 4) **Implementation details:** USV vs. UV, loss masking, and dropout interact non-trivially; early ablations were not always controlled for all three simultaneously.

C. Future Directions

- Train H-ViT on ImageNet-scale data at $N \geq 2048$.
- Develop efficient GPU kernels for H-matrix operations.
- Explore adaptive rank selection across H-matrix levels.
- Extend to larger LLMs (e.g., Llama-2 7B) where large d favours H-matrix structure.
- Continue multi-epoch experiments to fully characterise convergence behaviour.

VI. CONCLUSION

We have presented a systematic investigation of H-matrix approximations as structured replacements for dense transformer weights. In the ViT setting, H-matrix QKV projections reduce parameters by 48% and training time by 15% at $N=4096$, with only $\approx 2\%$ accuracy loss on CIFAR-10. In the LLM fine-tuning setting, an initial direct-replacement approach on WikiText-2 showed H-MAT competitive with LoRA, motivating a pivot to the additive adapter formulation $W = W_0 + \Delta W_{\mathcal{H}}$. Properly aligned H-matrix adapters (UV decomposition, dropout=0.05) converge to near-identical performance with LoRA after 3 epochs on medical QA, and come within 1.5 percentage points of LoRA on the CaseHOLD legal benchmark (78.7% vs. 80.2%), both representing absolute gains of over 55 points above the pretrained baseline. Across two distinct specialised domains, H-MAT consistently closes to within $\approx 1\text{--}2\%$ of LoRA at matched parameter counts. The key practical barrier—GPU inefficiency at small N —diminishes as model scale grows, positioning H-matrix adapters as a promising direction for very large models where the hidden dimension naturally exceeds current BLAS efficiency thresholds.

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